Investigating the Effects of a Subtropical Stratocumulus Cloud Breakup in Warm Climates Using Cloud-Locking Experiments

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ABSTRACT: In the warm, equable climate of the Eocene, constraints on CO_2 levels and low-latitude temperatures have generally precluded climate models from recreating key features of the climate, especially the above-freezing winter temperatures in the continental interiors suggested by fossil evidence of frost-intolerant species at high latitudes. Several cloud feedbacks have been suggested as mechanisms for enhanced wintertime warming, including a breakup of subtropical stratocumulus cloud decks at high CO_2 concentrations. It has been suggested that this breakup could lead to 8 K of global average warming, but how this low-latitude cloud feedback translates to high-latitude warming is not obvious and warrants quantification with a global climate model. In this study, we use cloud-locking experiments in the Community Earth System Model, version 2 (CESM2), in which homogeneous subtropical stratocumulus clouds are prescribed or completely removed to investigate the maximum warming achieved by a breakup of subtropical stratocumulus clouds in both a preindustrial and a high CO_2 climate. We use the present-day continental configuration and vegetation to make these results applicable to a future warm climate. We find that in the most dramatic case, the stratocumulus breakup leads to a significant globally averaged warming of about 4.5 K and contributes to some reduction in below-freezing days in the continental interiors at high latitudes. The resulting warming is limited by stabilizing low-cloud feedbacks induced elsewhere following the breakup. We conclude that a stratocumulus breakup may have played a nonnegligible role in past warm climates, even if it cannot, on its own, explain the key features of the Eocene.

KEYWORDS: Cloud forcing; Stratiform clouds; Paleoclimate

1. Introduction

During the early Eocene climate optimum (EECO, 53.3–49.1 Myr ago), the global average surface temperature was about 26°C and the latitudinal temperature gradient was at least 30% smaller than at the present (Lunt et al. 2021). Atmospheric CO_2 values are estimated to have been between 1000 and 3000 ppm during this era (Anagnostou et al. 2016). Although modern global climate models (GCMs) have had success in recreating some global features of this warm climate (Huber and Caballero 2011; Zhu et al. 2019), they have difficulties in producing strong enough wintertime warming in continental interiors (Eberle et al. 2010). Fossil evidence suggests the presence of frost-intolerant species such as palms and crocodiles at high latitudes (~50°N), near modern-day Wyoming (Greenwood and Wing 1995), far from the moderating effect of the ocean, where present-day winter temperatures often reach −40°C. These species could not have survived even a brief period below freezing, which necessitates a climate where no extreme cold events occur in this region for very long periods at a time. The seasonality problem, therefore, requires feedbacks that may not be well represented in current GCMs (Henry and Vallis 2022).

A number of hypotheses have been advanced to explain the Eocene climate, including an equator-to-pole Hadley cell (Farrell 1990) and strong ocean mixing driven by strengthened hurricanes (Emanuel 2002). Cloud feedbacks could provide additional warming mechanisms that may have been triggered during the Eocene at high CO_2, and multiple specific ideas were suggested in the context of the Eocene. These include polar stratospheric clouds (Sloan and Pollard 1998; Kirk-Davidoff et al. 2002; Dutta et al. 2023), a wintertime convective Arctic cloud feedback that produces strong warming and prevents the Arctic Ocean from freezing in the winter (Huber and Sloan 1999; Abbot and Tziperman 2008, 2009), low clouds over land due to the advection of moist air from the ocean which may prevent the surface from cooling to the freezing temperatures during winter (Cronin and Tziperman 2015; Hu et al. 2018), and subtropical stratocumulus cloud breakup increasing shortwave radiation and leading to warming (Schneider et al. 2019).

In this work, we focus on the role of the subtropical stratocumulus cloud feedback suggested by Schneider et al. (2019) in warm climates, such as the Eocene. Subtropical stratocumulus clouds currently cover about 6.5% of Earth's surface (Warren et al. 1988; Hahn and Warren 2007). These low-altitude clouds have a high albedo and can achieve a shortwave (SW) cloud radiative effect (CRE) of up to -100 W m^{-2}. These clouds form at the top of the atmospheric boundary layer below the strong trade wind inversion; strong cloud-top cooling drives turbulence

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that couples the cloud layer to the surface (Turton and Nicholls 1987; Bretherton and Wyant 1997; Wood 2012). Schneider et al. (2019) used a large-eddy simulation (LES) to show how increasing atmospheric CO_2 weakens the cloud-top cooling and can induce an abrupt breakup of the stratocumulus cloud deck. Salazar and Tziperman (2023) used a mixed-layer model to show that although the critical CO_2 value where this occurs is sensitive to both macrophysical and microphysical parameters such as SST, precipitation rates, and cloud-top entrainment efficiency, breakup may occur at CO_2 concentrations relevant for past warm climates such as the Eocene (Schneider et al. 2019).

As a first-order approximation for the warming a breakup of stratocumulus could cause, Schneider et al. (2019) suggested an elegant back-of-the-envelope calculation assuming a complete removal of stratocumulus clouds covering 6.5% of Earth's surface with a change in the SW CRE of 100 W m^{-2}. Schneider et al. (2019) used a climate sensitivity parameter representing "the more sensitive among current GCMs" to place an upper limit on the warming a stratocumulus breakup could cause. This specified sensitivity of 1.2 K $(\mathrm{W\ m^{-2}})^{-1}$ (or, its inverse, a global feedback parameter of 0.83 W m^{-2} K^{-1}) was used to calculate a globally averaged warming of up to 100 W m^{-2} × 6.5% × 1.2 K $(\mathrm{W\ m^{-2}})^{-1}$ = 7.8 K. As these clouds are concentrated at low latitudes, it is not immediately clear how the warming associated with a breakup of stratocumulus clouds would affect the higher latitudes and, in particular, the cold wintertime extremes in continental interiors. Thus, the relevance of such a breakup to the main open problem of the warm climate of the Eocene is not obvious. Additionally, using a climate sensitivity parameter that is specific to CO_2 forcing, as done above, assumes that the climate will respond to a localized forcing such as a stratocumulus breakup similarly to a spatially homogeneous forcing like CO_2. This is also not obvious and connects to previous work studying the SST pattern effect, which describes the dependence of climate sensitivity on the SST warming pattern (Langenbrunner 2020). Additionally, some previous work has examined the "radiative forcing pattern effect," which describes the dependence of the climate sensitivity on the meridional distribution of radiative forcing (Kang and Xie 2014; Zhang et al. 2023). This has direct relevance to studies of spatially heterogeneous aerosol forcing (Persad and Caldeira 2018; Persad 2022; Wilcox et al. 2023) and in particular marine cloud brightening (Latham et al. 2012; Stjern et al. 2018; Wan et al. 2024). The effect of a subtropical stratocumulus breakup, therefore, requires quantification using a global climate model, as we attempt to do here.

To explore the impact and global warming pattern associated with a breakup of subtropical stratocumulus clouds, we perform a series of cloud-locking experiments in the Community Earth System Model, version 2 (CESM2; Danabasoglu et al. 2020), at both preindustrial ($1xCO_2$) and high ($8xCO_2$) CO_2 concentrations. Although $8xCO_2$ is on the higher end of estimates of CO_2 levels during the Eocene, we choose such a high CO_2 level to make our results applicable to any generic warm climate and maximize the signal-to-noise ratio. We design the experiments to explore the most dramatic possible case: imposing completely homogeneous cloud decks in the subtropics with a SW CRE of about −100 W m^{-2} covering 6.5% of the globe and then completely eliminating them to study the resulting effects. This essentially recreates the abovementioned back-of-the-envelope calculation in a 3D model. Although this experiment is highly idealized, it allows us to set an upper bound on the warming associated with this cloud feedback.

In section 2, we provide a description of the cloud-locking method used in this study, with details of how we prescribed strong stratocumulus cloud decks. In section 3, we analyze the global and local effects of a subtropical stratocumulus breakup and compare the results to the expectation of the back-of-the-envelope calculation performed by Schneider et al. (2019). We also discuss the effect of a stratocumulus breakup on extreme cold events in the continental interiors of North America, given the observational constraints on such extremes during the Eocene. In section 5, we discuss the limitations of our approach and suggest future studies to more rigorously connect this work to the Eocene.

2. Methods

CESM2 is a fully coupled, state-of-the-art 3D global climate model with components representing the atmosphere, ocean, land, and sea ice (Danabasoglu et al. 2020). Compared to its predecessor CESM1, CESM2 shows an increase in equilibrium climate sensitivity (ECS) from 4 K to over 5 K, with much of this increase in climate sensitivity attributed to a weakening of negative cloud-phase feedbacks in the high latitudes, especially over the Southern Ocean (Gettelman et al. 2019; Bacmeister et al. 2020). In applications to paleoclimate, it has been demonstrated that this climate sensitivity may be too high as it simulates a Last Glacial Maximum (LGM) that is far too cold (Zhu et al. 2022). This motivated the development of a configuration of CESM2, which removes an inappropriate cap on cloud ice number and decreases the microphysics time step size (Zhu et al. 2022; PaleoCalibr). These changes allow CESM2 to simulate a more realistic LGM, with a lower ECS (~4 K) and weaker shortwave cloud feedbacks. PaleoCalibr also allows simulations at high CO_2 levels, which do not converge in the standard Community Atmospheric Model, version 6 (CAM6), configuration (Danabasoglu et al. 2020). For these reasons, we adopt the PaleoCalibr configuration for all of our simulations.

We run the model in a fully coupled configuration with nonactive ice sheets and a 1.9° × 2.5° horizontal atmospheric resolution. Following Zhu and Poulsen (2021), we run the model with a prescribed satellite-derived vegetation cover to exclude potential complications from the vegetation phenology feedback. We also assume modern Earth topography and continental distribution, as well as preindustrial aerosol emissions and non-CO_2 greenhouse gases. This allows us to isolate the joint effect of increased CO_2 and cloud forcing from continental geometry effects. While we contextualize this work with its relevance to the Eocene, we design our experiments such that the results are generalizable to a future warm climate as well. Future studies may implement Eocene boundary conditions as done, for example, by Zhu et al. (2019) in CESM1.

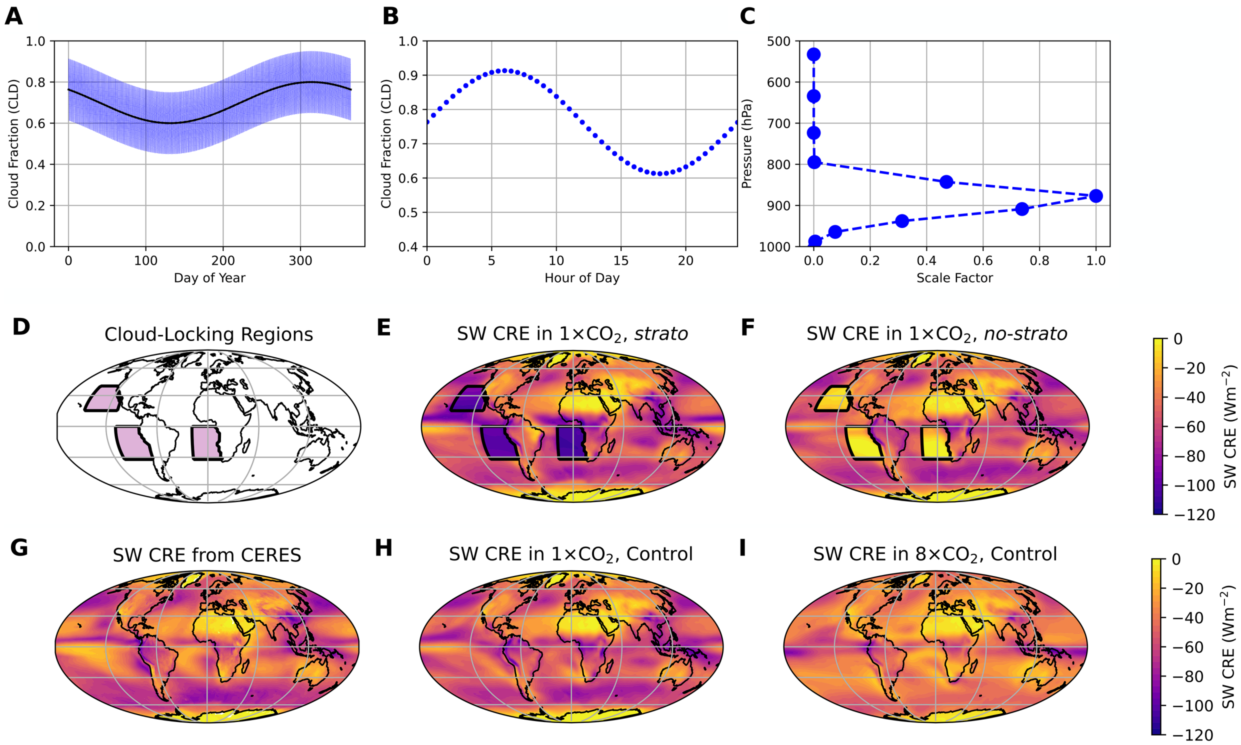

FIG. 1. Schematic of the regional cloud-locking experiment design, including the (a) seasonal and (b) diurnal cycles plus the vertical profile of the specified cloud properties (c). (d) Pink regions indicate where cloud-locking is applied. Shading demonstrates the SW CRE in (e) a "*strato*" run and a (f) "*no-strato*" run. (g) The SW CRE based on CERES observations (between July 2005 and June 2015) and (h),(i) the SW CRE output in the 1x and 8xCO_2 control simulations.

To investigate the maximum possible effect of a subtropical stratocumulus cloud breakup, we use the method of cloud locking. Cloud locking has been implemented in CESM2 to study the cloud radiative contribution to tropical precipitation extremes (Medeiros et al. 2021), Arctic amplification (Middlemas et al. 2020) (CESM1), El Niño–Southern Oscillation (Middlemas et al. 2019), the Madden–Julian oscillation (Benedict et al. 2020), and polar amplification in ice-free climates (England and Feldl 2024).

a. Cloud locking and cloud parameter specification

In a cloud-locking experiment, one prescribes the radiative effect of clouds, thus decoupling the cloud radiative effects from the instantaneous atmospheric state, possibly in a specified region and/or a limited range of altitudes. In CESM2, this requires specifying nine cloud parameters used in the radiation scheme: (i) the gridbox cloud fraction (model variable CLD), (ii) the cloud fraction adjusted for snow (CLDFSNOW), (iii) the in-cloud liquid water path (ICLWP), (iv) the in-cloud ice water path (ICIWP), (v) the in-cloud snow water path (ICSWP), (vi) the effective diameter for ice (DEI), (vii) the effective diameter for snow (DES), (viii) a size parameter for the gamma distribution of liquid water (MU), and (ix) a slope parameter for the gamma distribution of liquid water (LAMBDAC). These nine variables are specified at 2-h intervals throughout the simulation to capture the diurnal cycle of CRE. Because subtropical stratocumuli are liquid-phase clouds, we need only prescribed four out of nine cloud parameters that are used by the radiation scheme: (i) CLD, (ii) ICLWP, (iii) MU, and (iv) LAMBDAC, while the in-cloud ice water path is set to zero. Below, we describe the spatial, temporal, and vertical structure of these parameters.

In Fig. 1, we summarize the cloud-locking method used in this study. To most directly replicate the back-of-the-envelope calculation performed by Schneider et al. (2019), we wish to impose and remove stratocumulus clouds with a shortwave forcing of about -100 W m^{-2} over 6.5% of Earth. In previous applications of cloud locking, the cloud parameters were output from a control run and then used as inputs into a cloud-locked model run. However, as the SW CRE of subtropical stratocumulus is not uniformly -100 W m^{-2} in CESM2 (Figs. 1h,i) or observations (Fig. 1g), we do not extract the necessary cloud parameters directly from a control run of CESM2. Instead, we run the model with prescribed stratocumulus cloud values specified uniformly in specific regions of the modeling domain and then run it again with parameter values corresponding to the complete elimination of these clouds.

We center our cloud-locking regions on observed locations of high cover of subtropical stratocumulus clouds (Wood 2012): the northeast Pacific (NEP), southeast Pacific (SEP), and southeast Atlantic (SEA) (Fig. 1d, purple shading). The total surface area in which we apply cloud locking is 6.5% of the total Earth's surface area. Noting that subtropical stratocumuli are low-altitude clouds, we restrict the cloud locking to the bottom 10 atmospheric model layers so that all clouds

TABLE 1. Summary of prescribed parameters and values used for cloud locking. We also include the ranges of latitude $\Delta\phi$ and longitude $\Delta\theta$ for each of the three cloud-locking regions: the SEP, the SEA, and the NEP.

Cloud-locking parameters	Values
$\overline{\mathrm{CLD}}$	0.7
Δ^s_{CLD}	0.1
Δ^d_{CLD}	0.15
$\overline{\mathrm{ICLWP}}$	0.093 kg m^{-2}
$\Delta^s_{\mathrm{ICLWP}}$	0.013 kg m^{-2}
$\Delta^d_{\mathrm{ICLWP}}$	0.006 kg m^{-2}
n^s_{shift}	$40 \times 24 \times 2$
MU	6
LAMBDAC	3.2 m $\times 10^5$ m^{-1}
$\Delta\phi_{\mathrm{SEP}}$, $\Delta\theta_{\mathrm{SEP}}$	(33°S, 0°), (251°, 300°E)
$\Delta\phi_{\mathrm{SEA}}$, $\Delta\theta_{\mathrm{SEA}}$	(33°S, 0°), (341°, 20°E)
$\Delta\phi_{\mathrm{NEP}}$, $\Delta\theta_{\mathrm{NEP}}$	(15°, 40°N), (211°, 250°E)

above about 500 hPa are free to evolve and are radiatively coupled. This allows us to explore the response of high clouds in this area to a stratocumulus breakup. The latitude and longitude ranges defining the cloud-locking mask are listed in Table 1. As these are marine boundary layer clouds, we only cloud-lock over the ocean in grid boxes where the land fraction is less than 0.5.

We use analytic expressions for the seasonal and diurnal cycles of cloud fraction and liquid water path, in general agreement with observations (e.g., Ghate et al. 2009; Wood 2012), approximated as a sinusoid, and code these expressions into the radiation scheme. The evolution of cloud fraction with time $\mathrm{CLD}(n_{\mathrm{step}})$ is written as

$$\mathrm{CLD}(n_{\mathrm{step}}) = \overline{\mathrm{CLD}} - \Delta^s_{\mathrm{CLD}} \sin\left[\frac{2\pi(n_{\mathrm{step}} - n^s_{\mathrm{shift}})}{n^{\mathrm{year}}_{\mathrm{step}}}\right] + \mathrm{CLD}^d(n_{\mathrm{step}}), \tag{1}$$

$$\mathrm{CLD}^d(n_{\mathrm{step}}) = \Delta^d_{\mathrm{CLD}} \sin\left\{\frac{2\pi[n_{\mathrm{step}} - n^d_{\mathrm{shift}}(\theta)]}{n^{\mathrm{day}}_{\mathrm{step}}}\right\}, \tag{2}$$

where $\overline{\mathrm{CLD}}$ is the yearly mean cloud fraction, Δ^s_{CLD} is the amplitude of the seasonal cycle in cloud fraction, n_{step} is the step number in CAM6 (one step every 30 model minutes), n^s_{shift} is the shift in the seasonal cycle to yield a peak in October (Ghate et al. 2009), $n^{\mathrm{year}}_{\mathrm{step}}$ is the total number of time steps per model year, and $n^{\mathrm{day}}_{\mathrm{step}}$ is the number of time steps per model day. We superimpose a diurnal cycle onto the seasonal cycle (Figs. 1a,b), $\mathrm{CLD}^d(n_{\mathrm{step}})$ with amplitude Δ^d_{CLD}. The shift in the diurnal cycle $n^d_{\mathrm{shift}}(\theta)$ is a function of longitude such that CLD peaks around 0600 LT (Wood 2012). All parameters are defined in Table 1. Although the seasonal maximum slightly varies between the three regions considered, it still occurs between July and October in all sites, and we impose the same seasonal cycle on all three regions for simplicity.

The initial values of the cloud-locking parameters were chosen from model output in the $1\mathrm{xCO_2}$-control run in the SEP, where the model-predicted SW CRE is the highest (Fig. 1h). However, even when applied homogeneously across the cloud-locking regions, this did not result in a strong enough average SW CRE for our purposes. We then performed progressive offline scaling of $\overline{\mathrm{CLD}}$ and $\overline{\mathrm{ICLWP}}$ wherein we ran the model at $1\mathrm{xCO_2}$ with the strong prescribed clouds. After 1 year, we calculated the annual average SW CRE within the cloud-locking region. We repeated this process, increasing $\overline{\mathrm{CLD}}$ and $\overline{\mathrm{ICLWP}}$ by 5% between each iteration, until the SW CRE in the cloud-locking regions was -100 W m^{-2} in the $1\mathrm{xCO_2}$ case.

Prescribing cloud variables this way allows us to forgo the more computationally expensive method of reading in files containing the cloud variables at 2-h intervals and gives us the ability to tune the radiative forcing imposed in the cloud-locking regions. For the in-cloud liquid water path, ICLWP, we assume that the seasonal and diurnal cycles follow the same function as cloud fraction, with amplitudes defined in Table 1 and taken from the high SW CRE region of the SEP in the $1\mathrm{xCO_2}$-control run. We also set the microphysical parameters describing the size distribution of liquid droplets, MU and LAMBDAC, to constants taken from the $1\mathrm{xCO_2}$-control simulation. Although this is an oversimplified representation of the temporal variations in the cloud parameters, it allows us to prescribe the desired strong SW CRE of interest here.

Finally, we prescribe the vertical structure of the stratocumulus clouds to be equal to the time average of the normalized vertical structure taken from the SEP region in CAM6 (Fig. 1c). We normalize the vertical structure to have a maximum of one. We then scale the vertical profile by the temporal value of the cloud fraction and in-cloud liquid water path from Eq. (1) and apply the resulting user-created cloud-radiative properties to every grid cell in the cloud-locked region.

To initialize the two CO_2 states, we use a 500-yr run at preindustrial CO_2 (hereafter referred to as $1\mathrm{xCO_2}$-control), and we spin up a 1200-yr run at $8\mathrm{xCO_2}$ ($8\mathrm{xCO_2}$-control). We then perform four cloud-locking simulations:

- $1\mathrm{xCO_2}$, *strato*, cloud locking with subtropical stratocumulus clouds covering 6.5% of Earth's surface with an average shortwave cloud forcing of -100 W m^{-2} at $1 \times$ preindustrial CO_2 (280 ppm).
- $1\mathrm{xCO_2}$, *no-strato*, cloud locking with no subtropical stratocumulus clouds at $1 \times$ preindustrial CO_2 (280 ppm).
- $8\mathrm{xCO_2}$, *strato*, cloud locking with subtropical stratocumulus clouds covering 6.5% of Earth's surface with an average shortwave cloud forcing of -100 W m^{-2} at $8 \times$ preindustrial CO_2 (2240 ppm).
- $8\mathrm{xCO_2}$, *no-strato*, cloud locking with no subtropical stratocumulus clouds at $8 \times$ preindustrial CO_2 (2240 ppm).

Simulations with no stratocumulus clouds in the cloud-locking regions (*no-strato)* are performed by setting the cloud fraction to zero there. Because we are prescribing clouds in the *strato* cases that have a stronger SW CRE than the model-created clouds, we expect the *strato* simulation to be colder

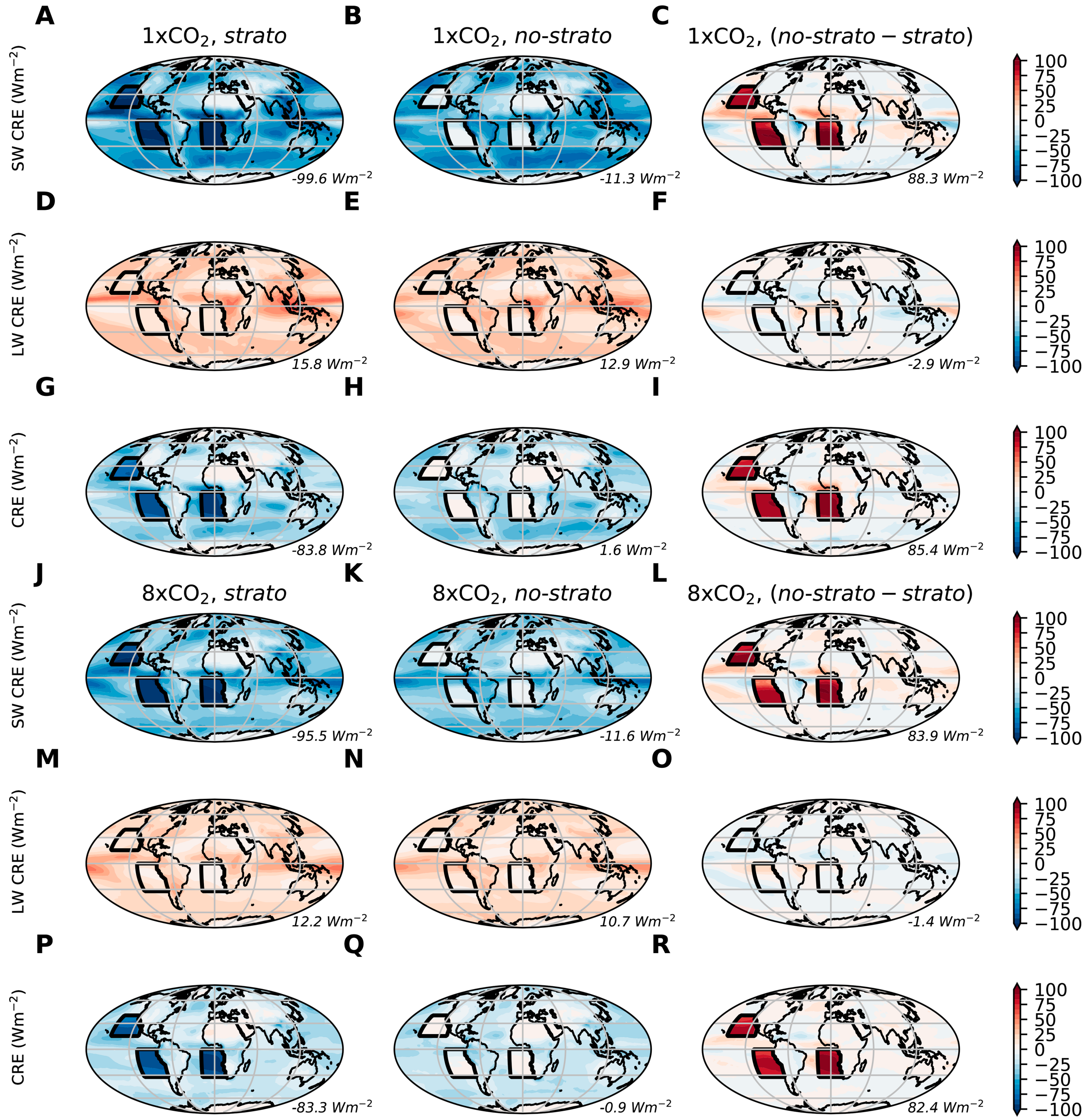

FIG. 2. (a)–(c),(j)–(l) SW CRE, (d)–(f),(m)–(o) LW CRE, and (g)–(i),(p)–(r) net CRE (SW + LW) in the last 30 years of the (a)–(i) $1xCO_2$ cloud-locking simulations and (j)–(r) the $8xCO_2$ cloud-locking simulations. Numbers in the lower right-hand corner of each subplot are the average values in the cloud-locking regions, indicated by the black contour. The cloud-locking parameters were tuned offline to yield an average SW CRE of approximately -100 W m^{-2} in the cloud-locking regions in (a) the $1xCO_2$ case. (i),(r) The influence of high clouds and the longwave forcing of the stratocumulus clouds decreased the intended total CRE to about 85 W m^{-2}. All CREs are evaluated at the top of the atmosphere.

than the control. Likewise, the *no-strato* cases are expected to be warmer than the control. We will, therefore, focus on the *difference* between the two cloud-locked simulations at each CO_2 level. Throughout, we will refer to the difference between the *no-strato* and *strato* simulations as the warming induced by stratocumulus cloud removal.

In Fig. 2, we demonstrate the difference in the CRE (LW CRE + SW CRE) between the *no-strato* and *strato* simulation in the $1xCO_2$ (Fig. 2i) and the $8xCO_2$ (Fig. 2r) cases in the last 30 years of the simulations. The offline tuning of the prescribed cloud properties creates a SW CRE in the cloud-locking regions that is about -100 W m^{-2} in the $1xCO_2$ (Fig. 2a).

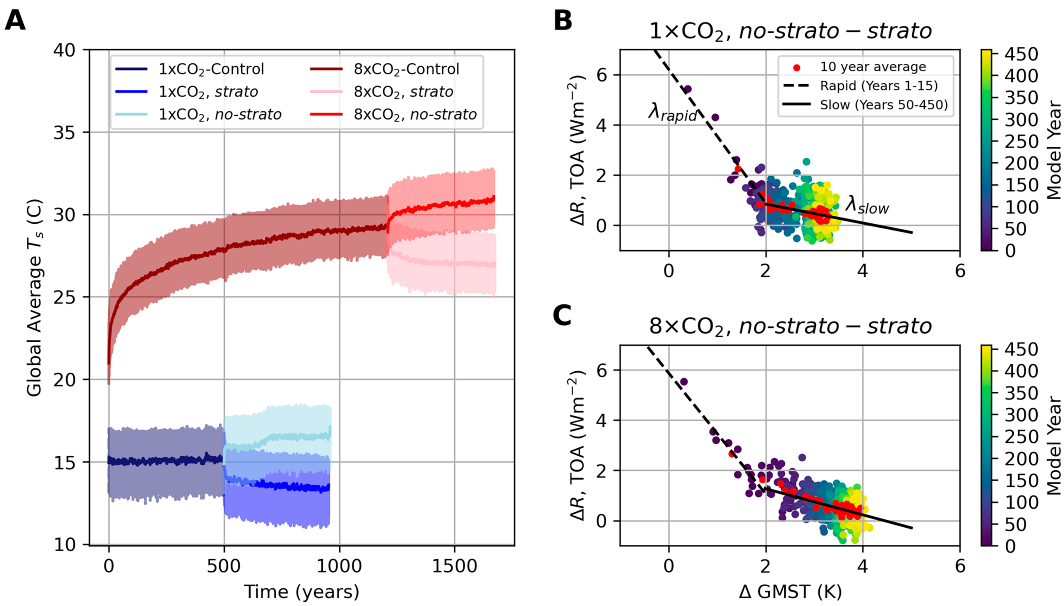

FIG. 3. (a) Time series of the six simulations explored here. The 8xCO_2 runs are plotted in red tones, while the 1xCO_2 runs are plotted in blue tones. The cloud-locked runs are branched from the corresponding control runs and then run for another 450 years. The solid lines show the yearly mean values, while the transparent lines (shading) show the corresponding monthly time series. (b) Difference in mean global TOA radiation imbalance ΔR between *no-strato* and *strato* in the 1xCO_2 case as a function of difference in the global mean surface temperature. The red dots denote the 10-yr averages. A linear extrapolation for years 1–15 (the rapid adjustment phase, dashed line with slope λ_{rapid}) and years 50–450 (the slow adjustment, solid line with slope λ_{slow}) is shown. (c) As in (b), but for the 8xCO_2 case.

However, because the cloud locking is only performed below 500 hPa, when the stratocumulus clouds are completely removed, the combination of preexisting high clouds and high clouds that respond to the removal of stratocumulus causes a CRE of -11 W m^{-2} in the cloud-locking regions (Fig. 2b). Considering the 3 W m^{-2} LW CRE of the prescribed clouds (Fig. 2f), the net difference in CRE between the *no-strato* and *strato* runs in the 1xCO_2 case is $100 - 11 - 3 = 86$ W m^{-2} (Fig. 2i). Interestingly, although identical parameters were used, the SW CRE in the 8xCO_2, *strato* run is -95.5 W m^{-2} (Fig. 2j), likely due to the differences in the adjustment of high cloud fraction in the cloud-locking region at higher CO_2. The net difference in the CRE in the 8xCO_2 case is 82.4 W m^{-2} (Fig. 2r). The additional forcing of the high clouds as well as the LW forcing of the prescribed stratocumulus clouds therefore decreased the intended forcing in the cloud-locking experiments. As the CRE shown in Fig. 2 is measured at the end of the simulations (and therefore after atmospheric adjustment has taken place), we will further quantify the estimated instantaneous forcing R_o of the stratocumulus removal using the Gregory method.

b. *Gregory method and linear extrapolations of feedbacks and warming*

In Fig. 3a, we plot the time series of the global mean surface temperature of the six simulations involved in this study, demonstrating how we branch the cloud-locking runs from the relevant 8xCO_2 or 1xCO_2 control runs. After 1200 years, the 8xCO_2-control run retains only a small top-of-atmosphere (TOA) radiative imbalance (~0.3 W m^{-2}); as our main goal is to compare the two cloud-locked runs, they are branched from the same spinup, and we believe the fact that the spinup is not fully equilibrated should not drastically affect our results. The 1xCO_2, *strato* and 1xCO_2, *no-strato* are branched from a preexisting equilibrated 500-yr run at preindustrial CO_2. Including cloud-locking experiments at preindustrial CO_2 allows us to explore the state dependency of the climate response to a stratocumulus breakup.

All four cloud-locked simulations are run for 450 years. We chose to end the simulations at that point, as after 450 years in the 8xCO_2, *no-strato* run, the model crashed due to a known and commonly occurring CESM2 instability at high CO_2 (Lunt et al. 2021), whose exact mechanism has not been fully explained. In Figs. 3b and 3c, we show the difference in net global TOA radiative imbalance between the *no-strato* and *strato* simulations as a function of difference in the global mean surface temperature.

As noted by previous studies (Held et al. 2010; Bacmeister et al. 2020), the response of the climate to radiative perturbation can be split into two time scales: a rapid adjustment phase corresponding to atmospheric adjustment before much surface warming has occurred and a slow adjustment phase corresponding to longer-term adjustments related to ocean temperatures. To investigate the climate response to a breakup of stratocumulus compared to CO_2 forcing, we perform linear fits of each phase of the climate response (Gregory et al. 2004). The slope of the rapid adjustment λ_{rapid} and the slow

TABLE 2. Summary of parameters from linear fits of the rapid (years 1–15) and slow (years 50–the end) adjustment periods of the experiments. The initial radiative perturbation R_o is the y intercept of the rapid adjustment phase, and the total change in the surface temperature ΔT_s is the x intercept of the slow adjustment phase. All global feedback parameter λ values have units of watts per square meter per Kelvin. The 1xCO_2 (*no-strato–strato*) and 8xCO_2 (*no-strato–strato*) experiments refer to the difference between the *no-strato* and *strato* cloud-locking runs for each CO_2 level. The values in parentheses indicate how the answer changes if the Gregory method is performed on each cloud-locking simulation separately, and then, the results are summed together.

Experiment	1xCO_2, (*no-strato–strato*)	8xCO_2, (*no-strato–strato*)	8xCO_2-control
R_o (W m^{-2})	6.21 (6.15)	5.88 (5.85)	12.9
ΔT_s (K)	4.23 (4.30)	4.44 (4.25)	15.91
$\lambda\left(-\frac{R_o}{\Delta T_s}\right)$	−1.47 (−1.43)	−1.32 (−1.37)	−0.81
λ_{rapid}	−2.66	−2.41	−1.06
$\lambda_{\mathrm{rapid}}^{\mathrm{SW}}$	−0.37	−0.28	0.75
$\lambda_{\mathrm{rapid}}^{\mathrm{LW}}$	−2.3	−2.12	−1.78
λ_{slow}	−0.38	−0.52	−0.44
$\lambda_{\mathrm{slow}}^{\mathrm{SW}}$	1.52	1.16	1.35
$\lambda_{\mathrm{slow}}^{\mathrm{LW}}$	−1.89	−1.68	−1.79

adjustment λ_{slow} phases will later provide insights into the differences in climate sensitivity and resulting warming between our cloud-locking simulations and the expectation from the back-of-the-envelope calculation. We also decompose each global feedback parameter (the slopes of these fits) into its shortwave and longwave components. This analysis is similar to that of Bacmeister et al. (2020), who used this method to study the role of Southern Ocean clouds in the increase in climate sensitivity between CESM1 and CESM2. We diagnose the switch from a rapid to a slow adjustment phase by calculating the second derivative of the surface temperature time series and finding an inflection at year 50. To ensure we are capturing only the rapid adjustment, we define the rapid adjustment period as years 1–15 and the slow adjustment as years 50–end. These fits are shown in Figs. 3b, 3c and Table 2.

Note that the commonly used Gregory et al. (2004) method usually plots TOA imbalance versus surface temperature change between a warming scenario and an equilibrated control run. We analyze the difference between the *no-strato* and *strato* cases because, specifically in the 8xCO_2 case, the cloud-locking simulations are still adjusting toward a steady state after 450 years of integration. This is especially evident in the time series of global mean surface temperature (GMST) in the 8xCO_2, *no-strato* case in Fig. 3a, where there still appears to be a warming trend. Since the focus of this study is the *difference* in GMST between the *no-strato* and *strato* cases, we analyze the difference between the cloud-locking cases to subtract out the residual TOA imbalance due to CO_2 forcing in the spinup. The analysis yields very similar results when we use the Gregory method on each cloud-locked simulation separately (Fig. S1, Table S1 in the online supplemental material) and then sum the results (see Table 2).

At the end of the 450 years, the residual TOA imbalance between the *no-strato* and *strato* simulations is 0.43 W m^{-2} in the 1xCO_2 case and 0.29 W m^{-2} in the 8xCO_2 case. To estimate the additional warming between *no-strato* and *strato* that further integration could reveal, we use the linear fit of the slow adjustment phase to estimate how much additional warming is expected to bring the relative difference in TOA imbalance down to 0 W m^{-2}. The linear extrapolation predicts about 1.12 K more expected warming between 1xCO_2 *no-strato* and *strato* and 0.46 K in the 8xCO_2 case had we run the models to equilibrium. As a caveat, we note that the 8xCO_2, *strato* case has a TOA imbalance and GMST that has plateaued over the last few decades of the run at about 0.3 W m^{-2} and 300 K, respectively. Therefore, our extrapolation to equilibrium using the Gregory method applied to the difference between the *no-strato* and *strato* run largely captures the trend from the 8xCO_2, *no-strato* run. This may introduce an error into our estimate of the additional warming expected. To bound this error, we note that the other three of our cloud-locking simulations have λ_{slow} between 0.3 and 0.5 W m^{-2} K^{-1}. This implies that the projected warming associated with the 0.3 W m^{-2} residual of the 8xCO_2, *strato* run ($0.3/\lambda_{\mathrm{slow}}$) may be in the range 0.6–1 K, assuming a similar long-term equilibration response. This range indicates that there is a 0.4-K uncertainty in the warming in the 8xCO_2 case, which is only a small correction to the 4-K difference from the expectation of the back-of-the-envelope calculation. Additionally, the majority of our discussion of climate sensitivity will be focused on the rapid adjustment phase, which has a clear trend in all simulations. We will further consider the radiative imbalance when we discuss our results, but we do not believe further integration would significantly change our conclusions.

In Fig. 2, we showed the net change in cloud forcing within the cloud-locking regions at the end of the simulation. However, this does not provide an accurate estimation of instantaneous forcing because high clouds were allowed to adjust within those regions. Therefore, to estimate the instantaneous radiative forcing induced by the removal of stratocumulus R_o, we calculate the y intercept of the rapid adjustment phase in the Gregory plot, giving ΔR when the global temperature change is 0 K, before any adjustment has occurred. This yields a globally averaged forcing of 6.2 W m^{-2} in the 1xCO_2 case and 5.9 W m^{-2} in the 8xCO_2 case. These values are slightly smaller than the intended 6.5 W m^{-2} to match the back-of-the-envelope calculation, and we will consider this difference in our discussion of globally averaged temperature change.

Finally, we perform an identical analysis on the Gregory plot of the 8xCO_2-control spinup (not shown). All parameters calculated from these linear fits (including the SW and LW components of the global feedback parameters in both the rapid and slow adjustment phases) are summarized in Table 2, including for the 8xCO_2-control spinup. We will return to these values in our discussion of globally averaged temperature change after a complete removal of subtropical stratocumulus in section 3.

3. Results

We now discuss the results of the cloud-locking simulations. We begin (section 3a) with an overview of the state of the climate at the end of the 8xCO_2-control spinup. We then report the global warming induced by a stratocumulus breakup, taken as the difference between the *no-strato* and *strato* cloud-locking runs (section 3b), and note the difference in climate sensitivity between CO_2 forcing and a localized forcing like a stratocumulus breakup. We analyze this difference by exploring the effect of the breakup on the ITCZ, latitudinal warming patterns, warming within the cloud-locking regions, regional patterns of warming, and the response of low clouds outside the cloud-locking regions (sections 3c–g). Finally, we connect our work to the Eocene equable climate period by discussing how the breakup of subtropical stratocumulus affects extreme cold wintertime interior continental temperatures in North America (section 3h).

a. Climate state in 8xCO_2-control

At the end of the 8xCO_2-control spinup period (1200 yr), all sea ice has melted, and the global average surface temperature is 29.2°C, consistent with marine and terrestrial proxies for the global mean surface temperature in the Eocene (Huber and Caballero 2011; Zhu et al. 2019) and 14.2°C warmer than the 1xCO_2-control. Additionally, the 8xCO_2-control simulation shows a large reduction in the low cloud fraction compared to the 1xCO_2-control, especially in the subtropical stratocumulus regions and the Southern Ocean (Fig. 1i vs Fig. 1h). This is consistent with other climate change studies using CESM2 (Bacmeister et al. 2020).

At the end of the spinup period, the 8xCO_2-control run retains a small TOA radiative imbalance of ~0.3 W m^{-2}. As described in section 2b, we use the Gregory method to estimate an additional equilibrium warming of about 1.7°C relative to the 1xCO_2-control (giving a total GMST of 30.9°C). To estimate the global feedback parameter λ due to CO_2 in the PaleoCalibr configuration of CESM2, we divide the instantaneous radiative forcing due to CO_2 (R_o = 12.9 W m^{-2}, Table 2) by the mean equilibrium warming (14.2 + 1.7 = 15.9 K), which yields $\lambda = -0.81$ W m^{-2} K^{-1}. Its inverse, the climate sensitivity parameter, is therefore 1.23 K (W m^{-2})$^{-1}$, essentially equivalent to the climate sensitivity parameter used in the back-of-the-envelope calculation. This model is therefore ideal for investigating how the climate response to a localized forcing like a stratocumulus breakup may differ from a more homogeneous forcing like CO_2.

b. Mean warming caused by stratocumulus breakup

Figures 4a and 4b show the zonally averaged change in the surface temperature following the removal of subtropical stratocumulus in the 1xCO_2 case (blue) and the 8xCO_2 case (red), over the entire globe (Fig. 4a) and over land only (Fig. 4b). We also plot the global distribution of warming in the 8xCO_2 case (Fig. 4c) and the 1xCO_2 case (Fig. 4d). The globally averaged temperature change between a strong subtropical stratocumulus state and a no subtropical stratocumulus state after 450 years is 3.1 K in the 1xCO_2 case and 30% higher (4.0 K) in the 8xCO_2. This is a very significant global warming effect of the removal of stratocumulus clouds in both cases, even if it is about half the warming predicted by the back-of-the-envelope calculation (Schneider et al. 2019).

Although at the end of the 450-yr simulation the 8xCO_2 case experienced significantly more warming than the 1xCO_2 case, the equilibrium response to the removal of stratocumulus clouds is very similar in the two cases. Based on the residual TOA radiative imbalance, we calculate an *equilibrium* mean global warming of 4.2 K (1xCO_2) and 4.4 K (8xCO_2) between the *no-strato* and *strato* runs (Table 2). To estimate the climate sensitivity parameter, we divide this equilibrated warming by the estimated radiative forcing R_o calculated in section 2b, giving 0.68 K(W m^{-2})$^{-1}$ (1xCO_2) and 0.75 K(W m^{-2})$^{-1}$ (8xCO_2), values that are the inverse of the global feedback parameter λ in Table 2. Therefore, we find a relatively small 10% increase in climate sensitivity to a stratocumulus breakup between the 1xCO_2 and 8xCO_2 cases.

Interestingly, even when extrapolated to equilibrium, the climate sensitivity parameter to a stratocumulus breakup is nearly *half* that of the climate sensitivity parameter to CO_2 forcing [0.68 or 0.75 K(W m^{-2})$^{-1}$ compared to 1.23 K(W m^{-2})$^{-1}$], which results in nearly half as much warming as expected from the back-of-the-envelope calculation. This suggests that the climate response to a stratocumulus breakup cannot be extrapolated from the climate sensitivity to CO_2, as these forcings are inherently different. This is consistent with recent work connecting SST warming patterns to changes in climate sensitivity (Langenbrunner 2020). Compared to CO_2 forcing, there must be some stabilizing feedback that occurs following the breakup of stratocumulus that mitigates surface warming.

Although our results mainly focus on the *difference* between the *no-strato* and *strato* cloud-locked simulations to quantify the maximum global warming achieved by the breakup, it is also interesting to note the warming observed between the *no-strato* and the control simulations. This gives an estimate of the warming induced by the breakup of the model-created subtropical stratocumulus. At the end of the 450-yr *no-strato* simulation, this model predicts a GMST increase of 1.4 and 1.7 K in the 1xCO_2 and 8xCO_2 case, respectively (Fig. S2, red lines). Using the Gregory method to extrapolate additional warming in equilibrium (and accounting for the residual TOA imbalance in the 8xCO_2-control case), we find a warming of

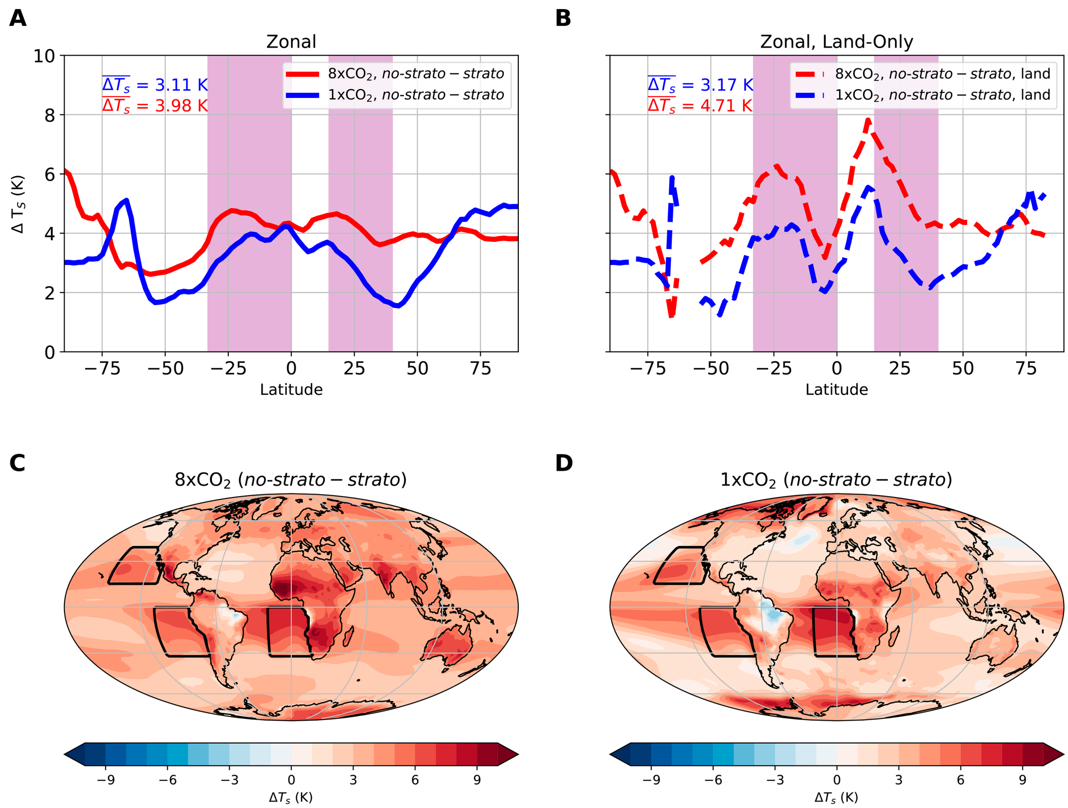

FIG. 4. (a) Difference in the zonally averaged surface temperature between the *no-strato* and *strato* cloud-locking simulations for the annual average over the entire globe; (b) the annual zonal average over land only. Purple shading indicates the latitude bands where cloud-locking is applied. (c) Difference in the surface temperature between the *no-strato* and *strato* cloud-locking simulations for annual mean in the $8xCO_2$ case, (d) annual mean in the $1xCO_2$ case.

1.65 K in the $1xCO_2$, *no-strato* case and 1.6 K in the $8xCO_2$, *no-strato* case. It is surprising that the $1xCO_2$ case experiences essentially the same warming as the $8xCO_2$ case following the removal of model-created clouds because the lower CO_2 has a higher stratocumulus cloud fraction in the control case than the $8xCO_2$ case (Fig. 1h vs Fig. 1i). This is also evidenced by the initial radiative forcing induced by the removal of the model-created clouds R_o in the *no-strato* case, where in the $1xCO_2$ case, $R_o = 3.3$ W m^{-2}, and in the $8xCO_2$ case, $R_o = 1.6$ W m^{-2} (Table S1). Strong negative feedbacks in the $1xCO_2$ case are therefore mitigating the surface warming compared to the control.

We now seek to understand the mechanism behind this negative feedback that is mitigating warming between the *no-strato* and *strato* cloud-locking runs. To do this, we investigate the pattern of surface warming at each CO_2 level and analyze changes to the ITCZ and cloud cover outside the cloud-locking regions.

c. *Effect of stratocumulus breakup on the ITCZ*

For both CO_2 values, between the *strato* and *no-strato* cloud-locking runs, there is a shift from a single ITCZ to a structure reminiscent of the double ITCZ often observed in GCMs (Zhang et al. 2019; Woelfle et al. 2019; Tian and Dong 2020). In the $8xCO_2$-control run, the model predicts a "double ITCZ," with bands of precipitation both north and south of the equator in the equatorial Pacific (Fig. S3). Previous work studying the cause of the "double-ITCZ problem" when simulating modern climate in GCMs attributed some of the effect to warm SST biases in the SEP due to insufficient marine stratocumulus cloud cover (Ma et al. 1996; Zhang et al. 2019; Woelfle et al. 2019). In CESM2, the east Pacific double-ITCZ bias was greatly improved by adjusting cloud microphysics parameters (which lead to reduced drizzle rates and rain-out of stratocumulus clouds) as well as tuning parameters in the CLUBB scheme (Woelfle et al. 2019). As subtropical stratocumulus clouds have been shown to greatly affect the ITCZ in models, it is no surprise that the strong stratocumulus-induced shortwave forcing imposed by our cloud-locking simulations drastically disrupts the ITCZ (Fig. S3). It is possible that in warm climates, the double ITCZ may be a physically realizable signal rather than a model bias. The change from a single to a double ITCZ between the *strato* and *no-strato* cases, for both CO_2 values, causes changes in the equatorial high cloud cover and precipitation.

d. *The latitudinal pattern of surface warming*

The latitudinal distribution of warming from stratocumulus removal is different between the $8xCO_2$ and $1xCO_2$ cases (Fig. 4a). In the former (red line), the warming is meridionally distributed quite uniformly, except for a small minimum in the Southern Ocean. In the 1xCO_2 case (blue line), warming is enhanced at low latitudes (say 25°S–25°N), but minima occur over the Southern Ocean, as in the 8xCO_2 case, and within the high-latitude Northern Hemisphere oceans. These minima also appear in the zonally averaged net surface shortwave (Fig. S4, yellow line). The warming signal in the Southern Ocean at both CO_2 levels may be damped by the negative cloud-phase feedback where ice droplets melt and become more reflective liquid droplets (Zhu and Poulsen 2020), which enhances the SW CRE there. This feedback is indicated by the negative change in net surface shortwave in the Southern Ocean, indicating an increase in cloud albedo (Figs. S5a,b). The lower background concentration of cloud ice in the 8xCO_2 case dampens this effect, but a decrease in SW CRE in the Southern Ocean is also observed in the higher CO_2 case. In the 1xCO_2 case, we additionally observe a notable reduction in net surface shortwave in the high-latitude Northern Hemisphere oceans (Fig. S5b). We will further investigate these warming minima in our discussion of cloud feedbacks in section 3g.

Warming due to the stratocumulus cloud removal is also enhanced in 1xCO_2 near the poles, where we find a smaller sea ice cover in the *no-strato* compared to the *strato* case, leading to large temperature differences. This sea ice effect does not occur in the 8xCO_2 case, as there is no sea ice in any of the 8xCO_2 runs.

e. Surface warming in cloud-locking regions

In the regions where the breakup is imposed (indicated by the black contours in Figs. 4c,d), the maximum local warming can be quite high, *up to* 8 K in both CO_2 regimes, close to the 9.5 K calculated by Schneider et al. (2019) within their LES domain. The *average* warming within our cloud-locking regions is approximately 5.5 K for both CO_2 values. This difference (9.5 vs 5.5 K) may be explained by the assumed fixed ocean heat uptake of -3 W m^{-2} in the surface energy budget of the slab ocean in the LES study of Schneider et al. (2019). In the cloud-locking regions in our study, the model calculates a significantly larger mean ocean heat uptake. The uptake, calculated as the difference in net surface heat flux between the *no-strato* and *strato* runs (Figs. S5i,j), is about -13 W m^{-2}, where negative values correspond to residual heat that is absorbed by the ocean.

f. The regional patterns of surface warming

Over land (Fig. 4b), the removal of stratocumulus in both CO_2 states results in peaks in warming at low latitudes. There is a strong warming signal specifically over land regions at latitudes of 10°–25° in both hemispheres. In the 1xCO_2 case, there are strong peaks in warming specifically in the Sahel and in southeastern Africa as well as smaller signals in India, Australia, and southern Mexico. In all five regions, local decreases in precipitation (Fig. S6f) lead to a reduction in soil moisture (Fig. S6d) and a corresponding decrease in latent heat flux, and therefore evaporative cooling, from the surface (Fig. S6h). Regions that were not very arid to begin with, such as India, Mexico, and southeastern Africa (Fig. S6b), experience a huge reduction in soil moisture following a stratocumulus breakup.

In the 8xCO_2 case, the warming signal displays similar patterns: the Sahel, Mexico, Australia, and India experience a reduction in soil moisture (Fig. S6c) due to the decreases in precipitation (Fig. S6e) in these regions, which weakens evaporative cooling (Fig. S6g) and leads to enhanced warming. Interestingly, there is also a strong warming signal over Southern Africa, which does not appear to experience substantial drying (Fig. S6c). Although the soil does not dry much following a stratocumulus breakup, we see that soil moisture is already extremely low in this region in the 8xCO_2-control case (Fig. S6a). Therefore, when heat from the cloud-locking regions is transported to this region, the low soil moisture limits the ability of evaporative cooling to mitigate surface warming due to the stratocumulus breakup (Zhou 2016; Duan et al. 2020). Other regions with low soil moisture such as Saudi Arabia also experience a similar warming signal.

A notable difference in the temperature change following stratocumulus removal in the two CO_2-states occurs in the North Atlantic, where the 1xCO_2 case experiences cooling following a stratocumulus breakup (Fig. 4d), while the 8xCO_2 experiences warming (Fig. 4c). In the 1xCO_2 case, the warming associated with the removal of stratocumulus clouds induces a transient weakening of the Atlantic meridional overturning circulation (AMOC; Fig. S7) following the melting of Arctic sea ice similar to that found, for example, by Nobre et al. (2023) in an abrupt CO_2 quadrupling experiment. The weakening occurs rapidly following the stratocumulus breakup and persists for around 200 years before partially recovering within 200 years. At the end of the 450-yr run, AMOC in 1xCO_2, *no-strato* is still about 15% weaker than in 1xCO_2, *strato* leading to the surface temperature signal seen in Fig. 4d. It is possible that this signal may disappear if the model was run to a steady state. In the 8xCO_2 case, AMOC is very sluggish to begin with, with a circulation strength about half of that in the 1xCO_2-control run. The cloud locking (both *strato* and *no-strato*) does not lead to further weakening in this case. The very weak AMOC in the high CO_2 case may be related to the nonequilibration of the 8xCO_2-control, specifically in the deep ocean, so interpreting AMOC-related responses is difficult here and outside the scope of this study.

On the other hand, we find that the warming associated with a breakup of subtropical stratocumulus clouds in the 8xCO_2 case leads to a large reduction in the low cloud fraction in the North Atlantic (Fig. 5h), which leads to warming there. We investigate the nonlocal response of low clouds to stratocumulus breakup in the following section.

g. Nonlocal response of low clouds

In our analysis, we have noted regions with minima in warming associated with decreases in net surface shortwave. To further understand the local and global temperature change, we seek to understand how nonlocal feedbacks respond to the forcing due to the removal of stratocumulus. As noted previously, the climate sensitivity parameter to the

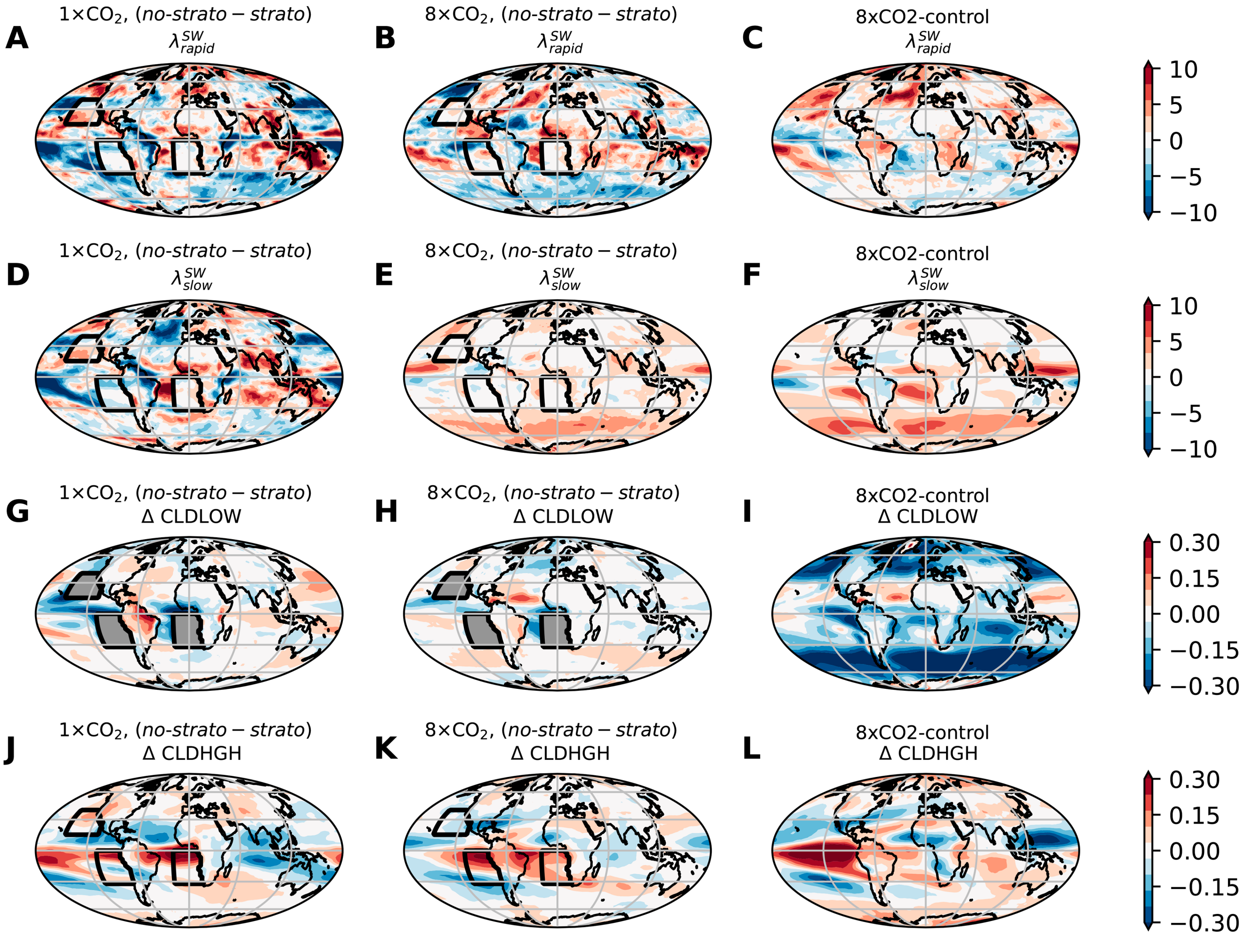

FIG. 5. (a)–(f) Slopes of the linear regression of the TOA net SW with respect to globally averaged change in surface temperatures as functions of latitude and longitude in the first 15 years [(a)–(c) rapid adjustment] and years 50–end [(d)–(f) slow adjustment]. The global average of each colormap represents the global SW feedback parameter for each period, as reported in Table 2. Negative (blue) colors indicate stabilizing feedbacks. These are associated with changes in (g)–(i) low and (j)–(l) high cloud cover at the end of each simulation. (top) The difference between *no-strato* and *strato* in the 1xCO_2 case. (middle) As in (top), but for the 8xCO_2 case. (bottom) The high CO_2 spinup case. Within cloud-locking regions (indicated by black contours), the change in low cloud is grayed out, as the model-created clouds there are ignored by the radiation calculation.

removal of stratocumulus is nearly half of that to CO_2 forcing in the 1xCO_2 and 8xCO_2 cases compared to that of CO_2 forcing. Therefore, it is clear that the climate response to a localized forcing like that of a subtropical stratocumulus breakup is fundamentally different than the climate response to a more homogeneous forcing like CO_2.

To further quantify this effect, we examine the global feedback parameters for each set of experiments (λ values in Table 2). We observe that the global feedback parameters in the rapid adjustment phase (λ_{rapid}, the slope of TOA imbalance versus surface temperature in the first 15 years) between forcing due to stratocumulus removal [the 1xCO_2, (*no-strato* − *strato*) and 8xCO_2-(*no-strato* − *strato*) cases] are over twice as large as for the forcing due to CO_2 (the 8xCO_2-control). This indicates that during the rapid adjustment phase following a stratocumulus breakup, strong negative feedbacks are acting, which reduce the TOA radiative imbalance without inducing much surface warming. Decomposing λ_{rapid} into its shortwave and longwave components provides a clue for which process is providing this negative feedback, which is mitigating surface warming.

Interestingly, during the rapid adjustment phase following a stratocumulus breakup, shortwave feedbacks are *negative* (negative $\lambda^{\mathrm{SW}}_{\mathrm{rapid}}$), opposite from the response to CO_2 forcing, which experiences positive $\lambda^{\mathrm{SW}}_{\mathrm{rapid}}$. Understanding the mechanism behind this difference in shortwave feedbacks between a stratocumulus breakup and a CO_2 forcing is therefore crucial to understanding why the cloud-locked simulations experienced 4.5 K of warming instead of the expected 8 K.

In Fig. 5, we plot the spatial pattern of the rapid and slow shortwave feedback parameters (calculated as the slope of the linear regression of net local TOA shortwave radiation with GMST) as well as changes in high and low clouds for both cloud-locking simulations and the 8xCO_2-control spinup. This will allow us to compare the response of clouds and therefore shortwave feedbacks between a stratocumulus breakup and

CO_2 forcing. As Figs. 5a–f are plotting the *slope* of local TOA shortwave forcing with global surface temperature change and the radiative forcing we impose in the cloud-locking regions is constant in time, the direct radiative forcing of low clouds in the cloud-locking regions does not contribute to the global feedback parameter. However, changes due to high clouds, which we allow to evolve freely, can be seen.

Both cloud-locking experiments show widespread negative (stabilizing, blue contours) shortwave feedbacks in the rapid adjustment phase (Figs. 5a,b), especially in the Southern Ocean and the equatorial Pacific. The 1xCO_2 case has additional negative shortwave effects in the midlatitude Pacific, while the 8xCO_2 case has a strong negative shortwave effect in the high-latitude eastern Pacific. Conversely, in the 8xCO_2-control (Fig. 5c) case, the rapid adjustment phase is marked by positive shortwave feedbacks, especially in the high-latitude Northern Hemisphere. This is reflected in the difference in sign of $\lambda_{\mathrm{rapid}}^{\mathrm{SW}}$ in Table 2 between the cloud-locked simulations and the 8xCO_2-control spinup run.

The shortwave feedbacks in the slow adjustment phase are more similar between the cloud-locking and CO_2-forced experiments (Figs. 5d,e). All three experiments have predominantly positive shortwave feedbacks, except in the equatorial Pacific, where the disruption of the ITCZ creates high clouds (Figs. 5j,k). The Southern Ocean exhibits a strong positive shortwave feedback in the 8xCO_2-control case and a somewhat less strong positive feedback in the 8xCO_2-locked case. This feedback does not appear very strongly in the 1xCO_2-locked case, perhaps due to the stronger negative phase feedback at lower CO_2 in the Southern Ocean (Zhu and Poulsen 2020). Finally, for CO_2 forcing, the subtropical stratocumulus regions have a strong positive shortwave feedback, consistent with the reduction in the low cloud fraction there. This positive stratocumulus feedback (e.g., change in TOA SW during the slow adjustment to steady state, divided by GMST change) does not appear in the cloud-locking regions because by design, we are fixing the clouds there and therefore also the TOA SW (apart from the small effect due to higher clouds). On the other hand, when increasing CO_2 without cloud locking, these regions contribute to the global feedback parameter because stratocumulus clouds do change there. This may contribute to the lower climate sensitivity that we find to the removal of subtropical stratocumulus versus an increase in CO_2. Indeed, if we calculate the global feedback parameter during the slow adjustment phase of the CO_2-forced case (without cloud locking) while ignoring the stratocumulus locking regions defined in our study, we find a 10% reduction in λ_{slow}, indicating a slightly less sensitive climate. As a reminder, the climate sensitivity to a stratocumulus breakup in this study is *half* of that to CO_2 forcing, so this analysis demonstrates that this 50% difference is due to climate responses in other regions.

The difference in the sign of the shortwave feedbacks in the rapid adjustment phase between the cloud-locked simulations and the CO_2 forced case seems to play a key role in lower climate sensitivity to a stratocumulus breakup. To better understand this difference in sign, we plot the changes in the low cloud fraction (Figs. 5j–l) and high cloud fraction (Figs. 5g–i). The most obvious difference between the cloud-locked simulations and the CO_2 forcing is that the CO_2 forcing causes a widespread reduction in low clouds, especially in the Southern Ocean, the subtropics, and the Northern Hemisphere high latitudes. The cloud-locking simulations do predict a reduction in low clouds near the cloud-locking regions (blue signal in Figs. 5g,h), likely due to the high SST induced by the removal of stratocumulus there or advection of low clouds out of the cloud-locking regions in the *strato* case, but this effect is dampened by an increase in the low cloud fraction elsewhere. The increase in the low cloud fraction in the midlatitude Pacific in the 1xCO_2 case is consistent with the strong negative shortwave forcing seen in the rapid adjustment phase. These clouds form rapidly after the removal of the subtropical stratocumulus as the free troposphere warms due to the enhanced vigorous convection from the cloud-locking regions. This increases the lower-tropospheric stability (LTS) at higher latitudes (Fig. S8), as the SSTs there have not had time to warm up. As the ocean surface warms during the slow adjustment phase and LTS decreases (Fig. S8), some of these clouds dissipate (indicated by the positive shortwave feedback in the midlatitude Pacific in Fig. 5d), but at the end of the 450-yr simulation, there is still a net increase in the cloud fraction there, providing a stabilizing shortwave effect. This is consistent with the minima in warming in the Northern Hemisphere midlatitudes seen in Fig. 4.

The much-reduced baseline low cloud fraction in the 8xCO_2 cloud-locked case weakens this effect in the midlatitude Pacific, but there is a strong negative shortwave forcing in the high-latitude eastern Pacific corresponding to an increase in the low cloud fraction there. Additionally, in the Southern Ocean, a small increase in the low cloud fraction provides a negative shortwave feedback and may also contribute to the minima in warming observed there in Fig. 4. Changes in high clouds occur across the simulations due to the disruption of ITCZ and lead to stabilizing shortwave feedbacks in the equatorial Pacific, as the removal of stratocumulus (either via cloud locking or CO_2 forcing) promotes the formation of a double ITCZ. However, as these are high clouds, we find that when taking into account the longwave effect of the clouds, the net change in CRE due to the ITCZ shift is nearly zero.

This analysis demonstrates how climate sensitivity depends on the nature of the forcing. For homogeneous forcing like CO_2, positive shortwave feedbacks associated with the reduction of low clouds amplify warming, whereas the localized forcing of the stratocumulus removal induces the formation of low clouds elsewhere, which greatly mitigates warming.

h. *Effect of stratocumulus breakup on extreme cold winter temperatures*

So far, we have analyzed how the breakup of subtropical stratocumulus affects surface temperatures in a generic warm climate. We now connect our work to the Eocene, where the main challenge in explaining observations is the absence of subfreezing temperatures in continental interiors away from the moderating effect of the ocean, seen via fossils of

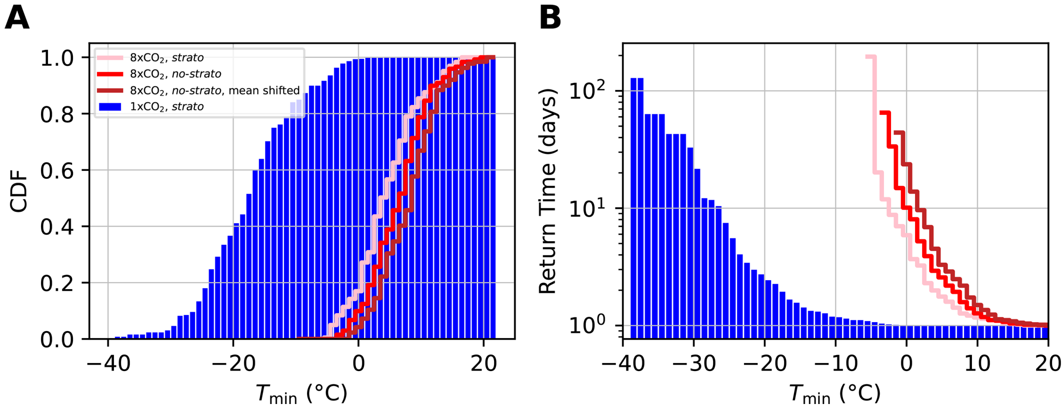

FIG. 6. (a) CDF of minimum daily temperatures in DJF in the continental interior of North America spanning (43°, 50°N) in latitude and (256°, 268°E) in longitude. (b) Return time in days of daily minimum temperatures. Below freezing days occur nearly daily in the 1xCO_2, *strato* case, about every 5 days in the 8xCO_2, *strato* case and every 10 days in the 8xCO_2, *no-strato* case. We also include a mean-shifted version of the 8xCO_2, *no-strato* case to compensate for the nonequilibration. The mean is shifted by 1.7 K for the nonequilibration of the 8xCO_2-control. Even in the 8xCO_2, *no-strato* case, there are still many below-freezing days per year in this region.

frost-intolerant plants and animals (Greenwood and Wing 1995). We investigate the role of a subtropical stratocumulus cloud breakup in mitigating extreme cold events in the continental interiors in Fig. 6. We output the minimum daily temperature in the last 50 years of each simulation in the region between 43°–50°N and 256°–268°E, near present-day Wyoming. As the temperatures of nearby grid cells are correlated, for each day during DJF, we select the coldest grid cell in this region such that we have 50 years × 90 days per winter = 4500 samples. The minima of these daily temperatures are shown in Fig. S9, where black contours indicate the region of interest.

In Fig. 6a, we plot the cumulative distribution function (CDF) for our coldest simulation (1xCO_2, *strato*, blue bars) and our warmest simulation (8xCO_2, *no-strato*, red line). We also show the CDF for the 8xCO_2, *strato* (pink line). In 1xCO_2, *strato*, nearly every winter day in this region experiences below-freezing temperatures. Increasing CO_2 (pink line) shifts the distribution such that the majority of winter days are above freezing; however, below-freezing days still occur. Removing subtropical stratocumulus (red line) further reduces, but does not eliminate, below-freezing days. We quantify this effect in Fig. 6b by showing how the return time of below-freezing days increases with CO_2 forcing and stratocumulus removal. In the 1xCO_2, *strato* simulation, the return time for below-freezing days is about 1 day, meaning essentially every day in DJF experienced below-freezing temperatures. Increasing CO_2 but holding the stratocumulus clouds fixed (8xCO_2, *strato*, pink line) increased the return time of freezing temperatures to 5 days, and removing the stratocumulus clouds further increased the return time to 10 days. To account for the nonequilibration of the 8xCO_2, *no-strato* simulation, we also provide a mean-shifted reference (brown line) in which we shifted the entire 8xCO_2, *no-strato* distribution by 1.7 K to account for the extrapolated additional warming of the 8xCO_2-control run. This uniformly shifted distribution represents the assumption that changes in extreme events are primarily driven by changes in the mean (Tziperman 2022). Even in the mean-shifted case, there are still a few days below freezing each winter (with a return time of 20 days).

The presence of frost-intolerant species suggests a return time of freezing events of thousands of years, if not longer. This is the time required for these species to remigrate northward following an event that led to their extinction. The increase in the return time from 5 to 20 days due to stratocumulus breakup indicates that although such a breakup can decrease the number of cold days in the high-latitude continental interiors, this effect is far from being enough on its own to explain the presence of frost-intolerant species there.

Additionally, the breakup of subtropical stratocumulus does not decrease the equator-to-pole temperature gradient as is expected for a warm climate such as the Eocene. As seen in Fig. 4a, in the 8xCO_2 case, the warming due to the breakup is distributed uniformly in latitude, except for the small minimum in the Southern Ocean. Therefore, this low-latitude cloud feedback does not translate to enhanced high-latitude warming in the configuration used here. The breakup of subtropical stratocumulus, though potentially relevant for the Eocene due to its effect on the global mean temperature, cannot on its own explain key features observed during this hothouse climate.

4. Discussion

The cloud-locking experiments performed here were highly idealized and meant to explore the *maximum* effect of a subtropical stratocumulus breakup. We prescribed clouds with a much higher shortwave forcing than the model-predicted clouds and those seen in observations, which was intended to set an upper bound on the globally averaged warming a breakup could induce. We also prescribed identical seasonal cycles in each of the three subtropical stratocumulus regions explored here, although in reality, their seasonal cycles can be offset by 2 or 3 months. Additionally, we note that although we ran the 8xCO_2-control simulation for 1200 years, the model was not fully equilibrated. Our use of the Gregory method on the difference between the *no-strato* and *strato* cases was designed to subtract out this residual, as both cloud-locking simulations were branched from the same nonequilibrated control. However, we acknowledge that residual warming, especially in the deep ocean, may slightly alter some details of our analysis. We believe our main conclusions are insensitive to this.

Additionally, we used a paleo-climate calibrated configuration of the model (PaleoCalibr), which allowed model convergence at high CO_2 but which has a lower (possibly more realistic) climate sensitivity than the standard CESM2 configuration (Zhu et al. 2022). A similar experiment using a model with a different climate sensitivity or different regional climate feedbacks may lead to different results. We also note that we used modern Earth topography, vegetation, and nonactive land ice in the cloud-locking experiments to isolate the effect of CO_2 and subtropical stratocumulus forcing from that of continental configuration and so that our results generalize to a future warm climate. A future study could impose Eocene boundary conditions for a more specific exploration of this feedback in the Eocene.

Finally, we used an extended version of the Gregory method in the analysis of negative cloud feedbacks following Bacmeister et al. (2020) by decomposing the TOA radiative imbalance into its SW and LW components and examining the response with warming during the rapid and slow adjustment periods. To explore the role of other feedbacks, radiative kernels could be used to decompose the global feedback parameter into its contributions from the water vapor feedback, the lapse rate feedback, the Plank feedback, the sea ice albedo feedback, and cloud feedback (Soden and Held 2006; Shell et al. 2008; Zelinka et al. 2012; Pendergrass et al. 2018) to further quantify the difference in the climate response to a CO_2 forcing and a stratocumulus breakup.

5. Conclusions

The breakup of subtropical stratocumulus clouds at high CO_2 has been suggested as a relevant feedback for hot house climates like the Eocene (Schneider et al. 2019). It was proposed using a back-of-the-envelope calculation that the removal of subtropical stratocumulus clouds with an average shortwave forcing of -100 W m^{-2} covering 6.5% of Earth's surface could induce a global mean warming of 8 K. In this work, we performed cloud-locking experiments in CESM2 at 1xCO_2 and 8xCO_2, wherein we prescribed clouds with an instantaneous mean shortwave forcing of about -100 W m^{-2} (the *strato* cloud-locking case) and completely removed low clouds in the cloud-locking regions (the *no-strato* case). We explain that this model experiment is designed to calculate the maximum possible global impact of the elimination of stratocumulus clouds.

We found that the globally averaged equilibrated warming from stratocumulus cloud removal was 4.2 K in the 1xCO_2 case and 4.4 K in the 8xCO_2 case. The model predicts a shift from a single to a structure similar to a double ITCZ between the *strato* and *no-strato* runs, consistent with previous work connecting the double ITCZ bias in modern climate simulations to cloud cover biases in the SEP. We also observed amplified regional warming in regions that experienced decreased precipitation or were already extremely dry such as the Sahel, Mexico, India, and Australia, indicating that local feedbacks can further amplify the effect of a remote stratocumulus removal.

While comparing the results of the cloud-locking runs to the expectation from the back-of-the-envelope calculation from Schneider et al. (2019), we noted that although the clouds we prescribed had an instantaneous shortwave forcing of about -100 W m^{-2}, multiple additional effects in the GCM in addition to those included in the back-of-the-envelope calculation reduced the global mean surface temperature response to half the value predicted in that calculation. This includes a small effect due to the presence of high clouds in these regions and the longwave effect of the removed stratocumulus clouds, which slightly decreased the intended forcing. The largest effect that we identified came from a difference in model climate sensitivity due to stabilizing low-cloud feedbacks at higher latitudes following the removal of the subtropical stratocumulus clouds. This stabilizing low-cloud feedback indicates that climate sensitivity parameters derived for more spatially homogeneous forcings like CO_2 cannot be used to accurately estimate warming due to localized forcing. This adds to the existing literature investigating the SST pattern effect (Langenbrunner 2020; Dong et al. 2020) and the radiative forcing pattern effect (Kang and Xie 2014; Zhang et al. 2023). This result also has relevance to studies of heterogeneous aerosol forcing (Persad and Caldeira 2018; Wilcox et al. 2023) and marine cloud brightening (Latham et al. 2012; Stjern et al. 2018; Wan et al. 2024).

We also connected our work to the main open problem of the Eocene equable climate period by examining how the breakup of subtropical stratocumulus affects extreme cold wintertime temperatures in the continental interior of North America. Fossil evidence of frost-intolerant species implies that temperatures in this region could not have been below freezing for even a brief period, and CO_2 forcing alone seems insufficient to completely mitigate these extreme events. We found that the breakup of subtropical stratocumulus at 8xCO_2 reduces the occurrence of below-freezing days from every 5 days to every 20 days. This is far from enough to explain the fossil observations, and, therefore, other climate feedbacks are likely necessary to explain the nonoccurrence of below-freezing days, such as the other proposed cloud feedbacks discussed in the introduction (Sloan and Pollard 1998; Kirk-Davidoff et al. 2002; Abbot and Tziperman 2008, 2009; Cronin and Tziperman 2015; Hu et al. 2018).

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Data availability statement. Simulations were performed using CESM2.1.4, which is freely available for download at https://github.com/ESCOMP/CESM. The dataset on which this paper is based is too large to be retained or publicly archived with available resources; however, all input files necessary to reproduce the simulations performed in this study as well as code to reproduce the figures can be found in the public GitHub repository: https://github.com/andreamsalazar/StratoCloudLocking_CESM2.

REFERENCES

Abbot, D. S., and E. Tziperman, 2008: A high-latitude convective cloud feedback and equable climates. *Quart. J. Roy. Meteor. Soc.*, **134**, 165–185, https://doi.org/10.1002/qj.211.

——, and ——, 2009: Controls on the activation and strength of a high-latitude convective cloud feedback. *J. Atmos. Sci.*, **66**, 519–529, https://doi.org/10.1175/2008JAS2840.1.

Anagnostou, E., and Coauthors, 2016: Changing atmospheric CO_2 concentration was the primary driver of early Cenozoic climate. *Nature*, **533**, 380–384, https://doi.org/10.1038/nature17423.

Bacmeister, J. T., and Coauthors, 2020: CO_2 increase experiments using the CESM: Relationship to climate sensitivity and comparison of CESM1 to CESM2. *J. Adv. Model. Earth Syst.*, **12**, e2020MS002120, https://doi.org/10.1029/2020MS002120.

Benedict, J. J., B. Medeiros, A. C. Clement, and J. G. Olson, 2020: Investigating the role of cloud-radiation interactions in subseasonal tropical disturbances. *Geophys. Res. Lett.*, **47**, e2019GL086817, https://doi.org/10.1029/2019GL086817.

Bretherton, C. S., and M. C. Wyant, 1997: Moisture transport, lower-tropospheric stability, and decoupling of cloud-topped boundary layers. *J. Atmos. Sci.*, **54**, 148–167, https://doi.org/10.1175/1520-0469(1997)054<0148:MTLTSA>2.0.CO;2.

Cronin, T. W., and E. Tziperman, 2015: Low clouds suppress Arctic air formation and amplify high-latitude continental winter warming. *Proc. Natl. Acad. Sci. USA*, **112**, 11 490–11 495, https://doi.org/10.1073/pnas.1510937112.

Danabasoglu, G., and Coauthors, 2020: The Community Earth System Model version 2 (CESM2). *J. Adv. Model. Earth Syst.*, **12**, e2019MS001916, https://doi.org/10.1029/2019MS001916.

Dong, Y., K. C. Armour, M. D. Zelinka, C. Proistosescu, D. S. Battisti, C. Zhou, and T. Andrews, 2020: Intermodel spread in the pattern effect and its contribution to climate sensitivity in CMIP5 and CMIP6 models. *J. Climate*, **33**, 7755–7775, https://doi.org/10.1175/JCLI-D-19-1011.1.

Duan, S. Q., K. L. Findell, and J. S. Wright, 2020: Three regimes of temperature distribution change over dry land, moist land, and oceanic surfaces. *Geophys. Res. Lett.*, **47**, e2020GL090997, https://doi.org/10.1029/2020GL090997.

Dutta, D., M. Jucker, S. C. Sherwood, K. J. Meissner, A. Sen Gupta, and J. Zhu, 2023: Early Eocene low orography and high methane enhance Arctic warming via polar stratospheric clouds. *Nat. Geosci.*, **16**, 1027–1032, https://doi.org/10.1038/s41561-023-01298-w.

Eberle, J. J., H. C. Fricke, J. D. Humphrey, L. Hackett, M. G. Newbrey, and J. H. Hutchison, 2010: Seasonal variability in Arctic temperatures during early Eocene time. *Earth Planet. Sci. Lett.*, **296**, 481–486, https://doi.org/10.1016/j.epsl.2010.06.005.

Emanuel, K., 2002: A simple model of multiple climate regimes. *J. Geophys. Res.*, **107**, 4007, https://doi.org/10.1029/2001JD001002.

England, M. R., and N. Feldl, 2024: Robust polar amplification in ice-free climates relies on ocean heat transport and cloud radiative effects. *J. Climate.*, **37**, 2179–2197, https://doi.org/10.1175/JCLI-D-23-0151.1.

Farrell, B. F., 1990: Equable climate dynamics. *J. Atmos. Sci.*, **47**, 2986–2995, https://doi.org/10.1175/1520-0469(1990)047<2986:ECD>2.0.CO;2.

Gettelman, A., and Coauthors, 2019: High climate sensitivity in the Community Earth System Model version 2 (CESM2). *Geophys. Res. Lett.*, **46**, 8329–8337, https://doi.org/10.1029/2019GL083978.

Ghate, V. P., B. A. Albrecht, C. W. Fairall, and R. A. Weller, 2009: Climatology of surface meteorology, surface fluxes, cloud fraction, and radiative forcing over the Southeast Pacific from buoy observations. *J. Climate*, **22**, 5527–5540, https://doi.org/10.1175/2009JCLI2961.1.

Greenwood, D. R., and S. L. Wing, 1995: Eocene continental climates and latitudinal temperature gradients. *Geology*, **23**, 1044–1048, https://doi.org/10.1130/0091-7613(1995)023<1044:ECCALT>2.3.CO;2.

Gregory, J. M., and Coauthors, 2004: A new method for diagnosing radiative forcing and climate sensitivity. *Geophys. Res. Lett.*, **31**, L03205, https://doi.org/10.1029/2003GL018747.

Hahn, C. J., and S. G. Warren, 2007: A gridded climatology of clouds over land (1971–96) and ocean (1954–97) from surface observations worldwide. Oak Ridge National Laboratory Tech Doc. ORNL/CDIAC-153 NDP-026E, 75 pp.

Held, I. M., M. Winton, K. Takahashi, T. Delworth, F. Zeng, and G. K. Vallis, 2010: Probing the fast and slow components of global warming by returning abruptly to preindustrial forcing. *J. Climate*, **23**, 2418–2427, https://doi.org/10.1175/2009JCLI3466.1.

Henry, M., and G. K. Vallis, 2022: Variations on a pathway to an early Eocene climate. *Paleoceanogr. Paleoclimatol.*, **37**, e2021PA004375, https://doi.org/10.1029/2021PA004375.

Hu, Z., T. W. Cronin, and E. Tziperman, 2018: Suppression of cold weather events over high-latitude continents in warm climates. *J. Climate*, **31**, 9625–9640, https://doi.org/10.1175/JCLI-D-18-0129.1.

Huber, M., and L. C. Sloan, 1999: Warm climate transitions: A general circulation modeling study of the late Paleocene thermal maximum (~56 Ma). *J. Geophys. Res.*, **104**, 16 633–16 655, https://doi.org/10.1029/1999JD900272.

——, and R. Caballero, 2011: The early Eocene equable climate problem revisited. *Climate Past*, **7**, 603–633, https://doi.org/10.5194/cp-7-603-2011.

Kang, S. M., and S.-P. Xie, 2014: Dependence of climate response on meridional structure of external thermal forcing. *J. Climate*, **27**, 5593–5600, https://doi.org/10.1175/JCLI-D-13-00622.1.

Kirk-Davidoff, D. B., D. P. Schrag, and J. G. Anderson, 2002: On the feedback of stratospheric clouds on polar climate. *Geophys. Res. Lett.*, **29**, 1556, https://doi.org/10.1029/2002GL014659.

Langenbrunner, B., 2020: The pattern effect and climate sensitivity. *Nat. Climate Change*, **10**, 977, https://doi.org/10.1038/s41558-020-00946-y.

Latham, J., and Coauthors, 2012: Marine cloud brightening. *Philos. Trans. Roy. Soc.*, **A370**, 4217–4262, https://doi.org/10.1098/rsta.2012.0086.

Lunt, D. J., and Coauthors, 2021: DeepMIP: Model intercomparison of early Eocene climatic optimum (EECO) large-scale climate features and comparison with proxy data. *Climate Past*, **17**, 203–227, https://doi.org/10.5194/cp-17-203-2021.

Ma, C.-C., C. R. Mechoso, A. W. Robertson, and A. Arakawa, 1996: Peruvian stratus clouds and the tropical Pacific circulation: A coupled ocean-atmosphere GCM study. *J. Climate*, **9**, 1635–1645, https://doi.org/10.1175/1520-0442(1996)009<1635:PSCATT>2.0.CO;2.

Medeiros, B., A. C. Clement, J. J. Benedict, and B. Zhang, 2021: Investigating the impact of cloud-radiative feedbacks on tropical precipitation extremes. *npj Climate Atmos. Sci.*, **4**, 18, https://doi.org/10.1038/s41612-021-00174-x.

Middlemas, E., A. Clement, and B. Medeiros, 2019: Contributions of atmospheric and oceanic feedbacks to subtropical northeastern sea surface temperature variability. *Climate Dyn.*, **53**, 6877–6890, https://doi.org/10.1007/s00382-019-04964-1.

Middlemas, E. A., J. E. Kay, B. M. Medeiros, and E. A. Maroon, 2020: Quantifying the influence of cloud radiative feedbacks on Arctic surface warming using cloud locking in an Earth system model. *Geophys. Res. Lett.*, **47**, e2020GL089207, https://doi.org/10.1029/2020GL089207.

Nobre, P., and Coauthors, 2023: AMOC decline and recovery in a warmer climate. *Sci. Rep.*, **13**, 15928, https://doi.org/10.1038/s41598-023-43143-5.

Pendergrass, A. G., A. Conley, and F. M. Vitt, 2018: Surface and top-of-atmosphere radiative feedback kernels for CESM-CAM5. *Earth Syst. Sci. Data*, **10**, 317–324, https://doi.org/10.5194/essd-10-317-2018.

Persad, G. G., 2022: The dependence of aerosols' global and local precipitation impacts on emitting region. *Atmos. Chem. Phys.*, **23**, 3435–3452, https://doi.org/10.5194/acp-23-3435-2023.

——, and K. Caldeira, 2018: Divergent global-scale temperature effects from identical aerosols emitted in different regions. *Nat. Commun.*, **9**, 3289, https://doi.org/10.1038/s41467-018-05838-6.

Salazar, A. M., and E. Tziperman, 2023: Exploring subtropical stratocumulus multiple equilibria using a mixed-layer model. *J. Climate*, **36**, 2421–2437, https://doi.org/10.1175/JCLI-D-22-0528.1.

Schneider, T., C. M. Kaul, and K. G. Pressel, 2019: Possible climate transitions from breakup of stratocumulus decks under greenhouse warming. *Nat. Geosci.*, **12**, 163–167, https://doi.org/10.1038/s41561-019-0310-1.

Shell, K. M., J. T. Kiehl, and C. A. Shields, 2008: Using the radiative kernel technique to calculate climate feedbacks in NCAR's Community Atmospheric Model. *J. Climate*, **21**, 2269–2282, https://doi.org/10.1175/2007JCLI2044.1.

Sloan, L. C., and D. Pollard, 1998: Polar stratospheric clouds: A high latitude warming mechanism in an ancient greenhouse world. *Geophys. Res. Lett.*, **25**, 3517–3520, https://doi.org/10.1029/98GL02492.

Soden, B. J., and I. M. Held, 2006: An assessment of climate feedbacks in coupled ocean–atmosphere models. *J. Climate*, **19**, 3354–3360, https://doi.org/10.1175/JCLI3799.1.

Stjern, C. W., and Coauthors, 2018: Response to marine cloud brightening in a multi-model ensemble. *Atmos. Chem. Phys.*, **18**, 621–634, https://doi.org/10.5194/acp-18-621-2018.

Tian, B., and X. Dong, 2020: The double-ITCZ bias in CMIP3, CMIP5, and CMIP6 models based on annual mean precipitation. *Geophys. Res. Lett.*, **47**, e2020GL087232, https://doi.org/10.1029/2020GL087232.

Turton, J. D., and S. Nicholls, 1987: A study of the diurnal variation of stratocumulus using a multiple mixed layer model. *Quart. J. Roy. Meteor. Soc.*, **113**, 969–1009, https://doi.org/10.1256/smsqj.47710.

Tziperman, E., 2022: *Global Warming Science: A Quantitative Introduction to Climate Change and its Consequences.* Princeton University Press, 366 pp.

Wan, J. S., C.-C. J. Chen, S. Tilmes, M. T. Luongo, J. H. Richter, and K. Ricke, 2024: Diminished efficacy of regional marine cloud brightening in a warmer world. *Nat. Climate Change*, **14**, 808–814, https://doi.org/10.1038/s41558-024-02046-7.

Warren, S. G., C. J. Hahn, J. London, R. M. Chervin, and R. L. Jenne, 1988: Global distribution of total cloud cover and cloud types over ocean. NCAR Tech. Note NCAR/TN-317+STR, 305 pp., https://www.osti.gov/servlets/purl/5415329.

Wilcox, L. J., and Coauthors, 2023: The Regional Aerosol Model Intercomparison Project (RAMIP). *Geosci. Model Dev.*, **16**, 4451–4479, https://doi.org/10.5194/gmd-16-4451-2023.

Woelfle, M. D., C. S. Bretherton, C. Hannay, and R. Neale, 2019: Evolution of the double-ITCZ bias through CESM2 development. *J. Adv. Model. Earth Syst.*, **11**, 1873–1893, https://doi.org/10.1029/2019MS001647.

Wood, R., 2012: Stratocumulus clouds. *Mon. Wea. Rev.*, **140**, 2373–2423, https://doi.org/10.1175/MWR-D-11-00121.1.

Zelinka, M. D., S. A. Klein, and D. L. Hartmann, 2012: Computing and partitioning cloud feedbacks using cloud property histograms. Part I: Cloud radiative kernels. *J. Climate*, **25**, 3715–3735, https://doi.org/10.1175/JCLI-D-11-00248.1.

Zhang, B., M. Zhao, H. He, B. J. Soden, Z. Tan, B. Xiang, and C. Wang, 2023: The dependence of climate sensitivity on the meridional distribution of radiative forcing. *Geophys. Res. Lett.*, **50**, e2023GL105492, https://doi.org/10.1029/2023GL105492.

Zhang, G. J., X. Song, and Y. Wang, 2019: The double ITCZ syndrome in GCMs: A coupled feedback problem among convection, clouds, atmospheric and ocean circulations. *Atmos. Res.*, **229**, 255–268, https://doi.org/10.1016/j.atmosres.2019.06.023.

Zhou, L., 2016: Desert amplification in a warming climate. *Sci. Rep.*, **6**, 31065, https://doi.org/10.1038/srep31065.

Zhu, J., and C. J. Poulsen, 2020: On the increase of climate sensitivity and cloud feedback with warming in the Community Atmosphere Models. *Geophys. Res. Lett.*, **47**, e2020GL089143, https://doi.org/10.1029/2020GL089143.

——, and ——, 2021: Last Glacial Maximum (LGM) climate forcing and ocean dynamical feedback and their implications for estimating climate sensitivity. *Climate Past*, **17**, 253–267, https://doi.org/10.5194/cp-17-253-2021.

——, ——, and J. E. Tierney, 2019: Simulation of Eocene extreme warmth and high climate sensitivity through cloud feedbacks. *Sci. Adv.*, **5**, eaax1874, https://doi.org/10.1126/sciadv.aax1874.

——, and Coauthors, 2022: LGM paleoclimate constraints inform cloud parameterizations and equilibrium climate sensitivity in CESM2. *J. Adv. Model. Earth Syst.*, **14**, e2021MS002776, https://doi.org/10.1029/2021MS002776.